

BRAGANCA, Portugal

WMO: 085750

Lat: 41.8037N

Lon: 6.7429W

Elev: 2270

StdP: 13.53

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.2	27.5	17.4	14.4	36.9	20.9	17.0	36.1	21.7	46.1	18.6	46.0	1.3	340	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.2	92.8	65.1	89.9	64.5	86.9	63.5	67.5	86.7	66.1	85.2	64.9	82.9	5.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.6	89.0	71.4	59.9	83.8	70.1	58.5	79.6	69.4	33.1	86.6	32.0	85.3	31.0	83.2	92.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	14.2	12.1	DB	20.8	97.4	2.7	1.8	18.8	98.7	17.2	99.8	15.6	100.8	13.6	102.1
			WB	19.9	70.9	2.9	4.7	17.9	74.3	16.2	77.0	14.6	79.6	12.5	83.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.2	41.2	43.4	48.8	52.0	58.1	66.3	71.1	71.0	64.7	56.7	46.8	41.9
	DBStd	12.18	5.47	5.10	5.73	6.47	6.78	6.84	5.95	5.74	5.78	5.65	5.56	5.44
	HDD50	1011	276	188	93	52	9	0	0	0	0	11	125	257
	HDD65	4168	738	604	504	393	231	67	16	14	75	263	545	718
	CDD50	2920	3	5	55	111	261	488	654	650	441	218	30	4
	CDD65	598	0	0	0	1	18	105	205	198	66	5	0	0
	CDH74	7462	0	0	2	45	283	1355	2483	2387	812	95	0	0
	CDH80	2950	0	0	0	2	61	509	1083	1040	240	15	0	0
(14) Wind	WSAvg	4.6	4.1	4.9	5.4	5.4	5.1	5.0	4.9	4.6	4.1	3.9	4.3	3.8
(15) Precipitation	PrecAvg	29.1	4.1	3.3	2.5	2.4	2.4	1.4	0.7	0.6	1.7	3.4	3.5	3.8
	PrecMax	45.4	15.1	11.8	12.9	6.7	5.6	4.6	4.8	3.4	5.9	9.8	9.6	13.7
	PrecMin	18.4	0.0	0.0	0.0	0.0	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.1
	PrecStd	7.1	3.1	2.7	2.5	1.6	1.4	1.1	0.9	0.7	1.5	2.4	2.6	3.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.8	64.5	72.3	78.7	86.1	94.0	96.0	97.3	91.5	81.9	65.6	57.4
		MCWB	49.2	50.5	52.7	57.7	62.9	65.7	66.3	65.8	63.5	58.6	55.2	51.5
	2%	DB	54.3	60.1	68.5	74.5	80.7	89.2	92.7	92.7	86.0	76.3	61.4	54.7
		MCWB	49.1	49.0	51.3	55.9	60.9	64.7	65.2	65.0	62.3	58.5	53.1	50.8
	5%	DB	52.1	56.5	65.2	70.6	76.8	85.8	89.8	89.5	82.2	71.6	58.3	52.5
		MCWB	48.4	47.8	50.5	54.6	59.2	63.5	64.8	64.0	60.7	57.5	52.3	48.9
	10%	DB	50.0	53.1	61.3	65.9	72.8	82.3	86.4	86.1	78.7	67.7	55.7	50.4
		MCWB	46.8	46.5	49.5	53.0	57.6	62.1	63.9	63.1	60.1	56.5	51.1	47.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	53.6	55.5	60.2	65.5	68.3	70.9	69.7	65.7	63.5	58.5	54.0
		MCDB	55.4	59.7	64.7	75.0	82.2	86.3	88.5	89.4	83.2	72.1	62.5	55.4
	2%	WB	51.0	51.3	53.5	57.7	62.6	66.6	67.8	66.9	63.9	61.3	55.6	51.9
		MCDB	52.9	56.8	63.0	71.5	76.9	84.7	86.8	87.6	80.9	69.7	58.7	54.1
	5%	WB	49.0	49.2	52.0	55.4	60.5	65.1	66.2	65.5	62.6	59.8	53.6	49.7
		MCDB	51.3	54.3	61.5	68.0	74.2	82.6	85.1	85.2	78.4	67.6	56.7	51.6
	10%	WB	47.1	47.4	50.5	53.4	58.6	63.5	64.8	64.2	61.1	58.2	51.9	47.9
		MCDB	49.5	52.3	59.2	63.7	70.5	79.6	83.1	82.2	74.7	65.2	55.2	49.9
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	17.4	19.6	19.6	21.3	24.3	26.2	26.8	23.5	18.7	15.0	13.4
		MCDBR	17.3	24.2	27.6	27.7	27.9	29.1	29.6	30.7	29.6	26.7	19.2	15.7
		MCWBR	12.2	15.0	14.5	13.1	12.7	11.6	10.9	11.0	11.8	12.8	12.2	11.2
	5% WB	MCDBR	12.2	19.6	22.8	25.2	25.4	27.3	28.0	28.3	25.4	19.8	14.3	11.4
		MCWBR	9.5	13.3	12.9	12.9	11.9	11.4	11.3	10.8	10.8	10.6	10.3	9.2
(40) Clear-Sky Solar Irradiance	taub		0.291	0.302	0.333	0.341	0.351	0.353	0.339	0.348	0.336	0.330	0.300	0.288
	taud		2.541	2.504	2.437	2.422	2.418	2.423	2.442	2.410	2.463	2.498	2.542	2.546
	Ebn at Noon		276	290	291	295	293	291	294	289	285	274	270	268
	Edh at Noon		22	27	32	35	36	36	35	35	31	27	22	20
(44) All-Sky Solar Radiation	RadAvg		559	882	1244	1600	1974	2268	2364	2089	1592	998	633	496
	RadStd		79	123	183	196	199	129	99	111	99	121	112	68

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+61

Nomenclature: See separate page